

## Research Statement<sup>\*</sup>

### 1 Overview

I am an econometrician whose research develops robust econometric and machine-learning methods for complex data. Much of my work begins from a simple empirical observation: the conditional mean is fundamental, but it does not always describe the feature of the distribution most relevant to the question being studied. Economic, financial, educational, public-health, and climate data may be asymmetric, heavy-tailed, multimodal, censored, spatially correlated, temporally dependent, or drawn from heterogeneous populations. Under these conditions, averages can be sensitive to extreme observations and can conceal dominant response patterns or other important distributional features. I therefore develop methods that target the conditional mode and related distributional objects. The conditional mode identifies the most likely response given covariates and is especially useful when interest centers on representative outcomes, local distributional shape, or robust prediction. Across these problems, I aim to preserve statistical targets while accommodating dependence, heterogeneity, incomplete observation, and scale. This perspective has led me from modal regression to modal volatility, forecast distributions, distributional dependence, and scalable learning procedures designed around clearly defined econometric targets.

My research agenda has three connected lines. *The first develops modal regression for empirical settings in which averages provide only a partial description of the conditional distribution.* Beginning with semi- and nonparametric mode estimation, I extend the framework to fixed-effects panels, varying-coefficient models, functional and semi-functional covariates, measurement error, random truncation, spatial dependence, and nonstationary regressors. These settings require different estimators and asymptotic arguments, but they share the same objective: making conditional-mode analysis feasible when data are nonlinear, incomplete, dependent, or otherwise nonstandard. *The second line studies dynamic economic environments, where persistence, volatility clustering, heavy tails, structural change, and evolving predictive relationships complicate conventional inference.* Applications in economics, finance, public health, and climate motivate this line because temporal dependence can alter both the location and shape of predictive distributions over time and horizons. I develop methods for nonlinear modal regression under dependence, modal and robust volatility modeling, online forecast combination, near-unit-root estimation, and distributional Granger causality. *The third line addresses the computational and statistical demands created by large, sequential, distributed, and heterogeneous data.* This work includes distributed mode learning, online and adaptive mode learning for streaming data, optimal subsampling for functional data,

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robust transfer learning, graph-based item response modeling, and distributional learning. Across these projects, machine-learning tools are used when they solve a specific problem of scale, information sharing, or data structure while preserving an explicit statistical target and a tractable basis for inference.

A unifying feature of this agenda is that the statistical target is chosen to match the empirical question rather than imposed by convention. In some applications, the relevant object is the most likely response; in others, it is volatility, a predictive distribution, or a distributional relationship between variables. This distinction matters when heterogeneity, tail behavior, dependence, or incomplete observation change what can be learned from the data. My work therefore develops estimators and inference procedures around each problem while retaining interpretable population targets. The resulting methods are designed to remain informative when a small number of observations can strongly affect averages, when dependence alters sampling behavior, or when the observed sample only partially represents the underlying population. This target-driven approach links my work across modal regression, time-series econometrics, and machine learning and provides a common basis for the specific contributions described below.

## 2 Modal Regression and Robust Nonparametric Econometrics

My first main research area develops modal regression as a general framework for robust semi- and non-parametric econometrics. The main econometric challenge is that the conditional mode is defined through the maximizer of a conditional density, so estimation requires recovering local distributional shape rather than only a conditional expectation. My work develops models, estimators, and asymptotic theory that make this target usable across flexible model structures, incomplete observation, dependence, and computational constraints that arise across modern empirical applications in economics and finance.

A first question in this line is how to retain flexible nonlinear structure without sacrificing interpretability or statistical efficiency. In *Semiparametric Partially Linear Varying Coefficient Modal Regression* (Ullah et al., 2023, **Journal of Econometrics**), we address the recurring tradeoff between the flexibility of fully nonparametric models and the interpretability of parametric specifications. We resolve this tension by combining linear components with varying-coefficient components under a modal regression objective, allowing some effects to remain directly interpretable while other effects vary flexibly with observed characteristics. A complementary strategy appears in *Nonparametric Estimator for Conditional Mode with Parametric Features* (Wang, 2024a, **Oxford Bulletin of Economics and Statistics**). The paper uses available parametric features to guide nonparametric mode estimation without changing the target parameter, showing how structural information can improve conditional mode estimation while preserving the robustness of the mode as the object of inference.

A second part of this research studies modal inference when the response distribution is only partially observed or when the covariate structure is more complex than in standard cross-sectional regression. These problems arise in two forms: the outcome may be incompletely observed, or the predictor itself may be a high-dimensional function or trajectory. In *Kernel Mode-Based Regression under Random Truncation* (Wang and Yao, 2026a, **Statistica Sinica**), the observed sample is distorted by random truncation, so the estimator must correct for the observation mechanism before recovering the conditional mode. In

*Robust Semi-Functional Censored Regression* (Wang, 2026, **Journal of Multivariate Analysis**), the response is censored and the covariates may be functional. The second challenge concerns complex predictors. In *Semi-Functional Varying Coefficient Mode-Based Regression* (Wang, 2025a, **Journal of Multivariate Analysis**), I study covariates represented by curves or trajectories rather than finite-dimensional variables. This setting is important for modern empirical work involving functional predictors such as time paths, signals, or repeated measurements. In *Optimal Subsampling for Functional Quasi-Mode Regression with Big Data* (Wang, 2025b, **Journal of Computational and Graphical Statistics**), I address the computational side of this problem by developing subsampling methods that retain the inferential target while reducing computational burden in large functional data sets.

A third part of this work addresses measurement error, spatial dependence, nonstationarity, and panel-data heterogeneity. Since modal regression depends on the conditional density, covariate measurement error can distort not only regression coefficients but also the estimated shape of the response distribution. In *Semiparametric Modal Regression with Varying Coefficients and Measurement Error* (Ullah and Wang, 2026, **Journal of Statistical Planning and Inference**), we study modal inference when covariates are contaminated. In *B-Splines Modal Estimation under Measurement Error with Deconvolution* (Wang, 2026a+, **The Canadian Journal of Statistics**), I develop spline-based deconvolution methods for modal estimation. A different complication arises when observations are dependent across locations rather than independently sampled. *Nonparametric Spatial Mode-Oriented Regression* (Wang and Yao, 2025a, **Electronic Journal of Statistics**) and *Nonparametric Spatial Modeling towards the Mode* (Wang and Yao, 2027, **Statistica Sinica**) develop modal methods that account for spatial dependence. Persistence in regressors creates another departure from standard asymptotic settings. *Kernel Mode-Based Varying Coefficient Models with Nonstationary Regressors* (Wang and Yao, 2026b, **Journal of Nonparametric Statistics**) studies this problem for mode-based varying-coefficient estimation, while *Adaptive Robust Estimation for Varying Coefficient Models* (Wang and Yao, 2026c, **Computational Statistics & Data Analysis**) develops adaptive robust procedures for varying-coefficient models. The same concern with heterogeneity appears in panel data, where unit-specific effects must be separated from the conditional modal relationship. In *Modal Regression for Fixed Effects Panel Data* (Ullah et al., 2021, **Empirical Economics**), we address this setting directly. Fixed-effects panel models control for unobserved heterogeneity but target conditional means. This paper develops a modal alternative that directly estimates the most likely conditional response while accounting for unit-specific effects.

This work extends modal inference across model structures, incomplete outcomes, functional covariates, measurement error, spatial dependence, persistent regressors, large data sets, and panel heterogeneity. Each setting requires different estimation or inference, but the conditional mode remains the common population object. These results make the conditional mode an estimable and inferentially tractable target across a broader class of econometric designs in empirical research.

### 3 Time Series and Distributional Dynamics

My second main research area develops robust methods for dependent and dynamic data. The central econometric difficulty is that serial dependence, persistence, heavy tails, volatility clustering, and struc-

tural change alter both the behavior of estimators and the predictive distributions they are intended to recover. These features are common in financial, macroeconomic, and climate data, where extreme shocks or changing dynamics can make average forecasts an incomplete description of future outcomes. My work addresses these problems through modal regression under dependence, robust volatility measures, distributional prediction, sequential forecast combination, and robust inference under high persistence.

A first question in this line is how modal regression can be estimated and justified when observations are serially dependent rather than independently sampled. In *Nonlinear Modal Regression for Dependent Data with Application for Predicting COVID-19* (Ullah et al., 2022, **Journal of the Royal Statistical Society Series A**), we develop nonlinear modal regression under dependence and use epidemic prediction to study the most likely conditional trajectory rather than only its average. This theoretical direction continues in *Nonlinear Kernel Mode-Based Regression for Dependent Data* (Wang, 2024b, **Journal of Time Series Analysis**), which develops kernel mode-based estimation under dependence conditions, and in *Parametric Modal Regression with Autocorrelated Error Process* (Wang, 2025c, **Statistica Sinica**), which studies modal regression when the regression errors are serially correlated. These papers develop modal estimation across nonlinear dependence, kernel-based settings, and serially correlated regression errors, where standard independent-sample arguments are unavailable.

Volatility modeling provides a natural application because financial time series are heavy-tailed and dominated by rare influential movements. Standard volatility methods summarize risk through conditional means of squared returns or related transformations, but such summaries can be sensitive to extremes. In *Modal Volatility Function* (Ullah and Wang, 2025, **Journal of Time Series Analysis**), we introduce a mode-based volatility function that characterizes the most likely conditional behavior of volatility-related outcomes. In *Mode Meets Mean: A New Robust Volatility* (Wang, 2025a+, **Journal of Time Series Analysis**), I connect modal and mean-based perspectives to construct a robust volatility measure. These papers develop volatility measures that retain information about dominant conditional behavior while reducing sensitivity to asymmetry, heavy tails, and unusually large movements in returns.

My more recent time-series work moves from point prediction to distributional prediction. *Testing Distributional Granger Causality with Entropic Optimal Transport* (Wang, 2026b+, **Journal of Time Series Analysis**) asks whether one time series helps predict the entire conditional distribution of another. Classical Granger causality is often formulated through conditional means, but predictive information may appear through variance, skewness, tail behavior, or other distributional features. Entropic optimal transport provides a computationally feasible way to compare conditional distributions while preserving a meaningful geometric interpretation. A related problem arises when several forecasting models must be combined while both dependence and the forecasting environment evolve over time. In *Online Randomized Distributionally Robust Forecast Combination for Dependent Data* (Wang, 2026c+, **Journal of Time Series Analysis**), I develop a sequential approach to this problem. The resulting procedure adapts to model uncertainty while explicitly accommodating dependent observations. High persistence creates a different inferential difficulty. In *On Robust Estimation for Moderate Deviations from a Unit Root* (Wang, 2026d+, **TEST**), I develop robust estimation for near-unit-root settings in which conventional procedures become sensitive to persistent dynamics and atypical observations.

This work develops time-series methods across serial dependence, nonlinear dynamics, volatility, dis-

tributional predictability, changing forecasting environments, and near-unit-root persistence. The statistical target differs across these problems, from conditional modes and volatility measures to entire predictive distributions, but each requires inference that accounts directly for the dynamic structure of the data. These results provide tools for studying how location, volatility, and distributional predictive relationships evolve over time rather than treating temporal dependence as a secondary complication.

## 4 Machine Learning and Scalable Econometrics

My third main research area connects robust econometric ideas with modern machine learning. The central challenge is that contemporary data may be large, high-dimensional, sequential, distributed, or heterogeneous, so classical nonparametric procedures can become computationally infeasible or statistically inefficient under privacy and distributional constraints. At the same time, predictive accuracy alone is not sufficient when the empirical question requires a clearly defined population target, interpretable structure, or valid statistical inference. My work in this area develops scalable learning procedures that preserve explicit statistical objects, such as conditional modes, transfer-learning targets, and distributional structure.

A first problem is how to preserve modal inference when data are distributed across machines or institutions rather than available in a single centralized sample. In *Distributed Learning for Kernel Mode-Based Regression* (Wang, 2025d, **The Canadian Journal of Statistics**), I develop distributed procedures for settings in which centralization is limited by scale, privacy, or institutional constraints. The method reduces computational burden while retaining the same conditional-mode target as centralized estimation. A related challenge arises when observations arrive sequentially rather than as a fixed sample. In *Online Kernel-Based Mode Learning* (Wang and Yao, 2025b, **Journal of Computational and Graphical Statistics**), we develop online updating procedures that incorporate new observations without repeated batch estimation while retaining the same population target. My more recent paper, *Adaptive Mode Learning for Nonstationary Streaming Data* (Wang, 2026e+, **Journal of Computational and Graphical Statistics**), extends this research to streaming environments in which the distribution may change as observations accumulate. These papers move from distributed computation to sequential updating and then to adaptive learning as the data-generating process evolves over time.

Transfer learning addresses a different problem, i.e., how to use information from related populations without importing bias from the source distribution. In economics and the social sciences, information from one market, region, institution, or population can improve estimation elsewhere, but differences between source and target distributions create the risk of negative transfer. In *Doubly Debiased Robust Subsampling for Transfer Learning* (Wang and Wong, 2026, **Journal of Machine Learning Research**), we develop subsampling and debiasing methods for transfer learning. The method therefore borrows source information only to the extent that it improves estimation for the target population while controlling transfer-induced bias. In *Geometry-Misalignment in Distributional Learning* (Wang and Zhong, 2026, **International Conference on Machine Learning**), we study a related foundational issue, i.e., the geometry imposed by a learning algorithm may not align with the statistical structure of the distributions being learned. These projects study when information can be shared across populations or representations without distorting the statistical structure that determines the target analysis.

An earlier application of this perspective appears in *A Machine Learning Strategy for Autism Screening in Toddlers* (Wang et al., 2019, **Journal of Developmental & Behavioral Pediatrics**), where machine-learning methods are used for a concrete screening decision. This application also reflects a theme that runs through my later work: computational methods should be shaped by the structure of the data and the statistical objective of the problem. The projects in this section address distributed storage, sequential arrival, nonstationarity, cross-population transfer, and distributional mismatch while preserving interpretable statistical targets. Across these settings, I adapt computational tools to the underlying econometric structure rather than treating predictive performance as the sole criterion for analysis.

## 5 Structured Econometric Applications

A further part of my research develops structured econometric methods for factor modeling, forecast evaluation, and educational measurement. Although the statistical objects differ across these settings, each problem requires structure beyond standard low-dimensional, mean-based, or conditionally independent formulations. The common question is how to extract useful structure from high-dimensional, heterogeneous, or networked data while retaining a clearly defined object for estimation and inference.

One problem is how to recover common structure from high-dimensional data when averages are distorted by heavy tails, asymmetry, or heterogeneous latent behavior. In *Mode-Adaptive Factor Models* (Wang, 2025e, **Scandinavian Journal of Statistics**), I develop a mode-oriented factor framework for this setting. The paper asks how common components can be estimated when the central tendency of the data is better represented by the most likely pattern than by the average pattern. The resulting estimator targets the dominant latent pattern directly, providing a mode-based approach to factor extraction in high-dimensional economic and financial data under heavy-tailed and asymmetric disturbances.

Forecast evaluation raises a different high-dimensional problem: the information used to assess rationality may generate many moment conditions simultaneously. In *Estimation and Testing of Forecast Rationality with Many Moments* (Lee and Wang, 2025, **Macroeconomic Dynamics**), we study forecast rationality when the information set generates many moment conditions. When the number of predictors, instruments, or implied restrictions is large, conventional low-dimensional forecast-rationality procedures no longer directly accommodate the available information set without discarding potentially informative restrictions. This paper develops methods for estimation and testing in such many-moment settings, contributing to the econometric literature on forecast evaluation and rational expectations.

Networked educational data create another form of structural complexity because responses may depend on relationships among students, items, or learning contexts in addition to individual latent traits. Item response models treat persons and items as conditionally independent after accounting for latent traits and covariates. In many learning environments, however, students, items, and response contexts are connected through networks. In *Graph Neural Item Response Model for Networked Learning Environments* (Wang and Zhong, 2026+, **Journal of Educational and Behavioral Statistics**), we incorporate graph structure into item response modeling while retaining interpretable psychometric components. The network information therefore augments the latent measurement structure rather than replacing it with an unrestricted prediction model.

These projects address three distinct forms of structural complexity: latent factors in high-dimensional data, many restrictions in forecast evaluation, and network dependence in educational measurement. The statistical targets differ, but each setting requires methods that use additional structure without sacrificing interpretability or tractable inference. This work extends the same target-driven approach underlying my modal and distributional research to problems where dimensionality, information sets, or network connections determine what can be learned from the data.

## 6 Future Research Agenda

My future research will build on this agenda by developing inference for distributional targets, changing environments, and structured information sharing. One direction is to deepen the theory of modal and distributional regression. Several inferential questions remain open, including simultaneous inference for modal functions, modal regression with many covariates, and causal designs in which treatment effects are represented by changes in the most likely outcome or the conditional distribution. I am especially interested in developing frameworks that compare and combine means, quantiles, modes, and optimal-transport-based summaries. A central question is when these different summaries lead to materially different empirical conclusions and how uncertainty can be quantified jointly across them.

A second direction is distributional time-series econometrics. A central problem is how predictive distributions should be estimated and compared when persistence, volatility, tails, and forecasting relationships change over time. My work on modal volatility, distributional Granger causality, online forecast combination, and near-unit-root estimation provides a starting point for developing procedures that adapt to these changes while retaining valid inference. I plan to study how distributional instability can be detected, how information should be discounted after structural change, and how predictive relationships can be updated without treating each new regime as an entirely separate problem.

A third direction is econometric machine learning with explicit targets. A central issue is how information from large, distributed, or related data sources should be used when their distributions differ from the target population or change over time. My work on distributed learning, nonstationary data streams, transfer learning, graph-based models, and distributional learning provides a foundation for studying when information can be shared safely and when borrowing should be reduced or stopped. I plan to develop adaptive information-sharing procedures that account explicitly for distribution shift, source-target mismatch, and dependence across evolving markets, institutions, and populations while preserving interpretable statistical targets and uncertainty quantification.

These directions will be developed through applications in financial volatility, economic forecasting, climate data, and educational measurement, where distribution shift, dependence, heterogeneity, and incomplete observation arise in different forms across samples, horizons, institutions, and populations. The applications provide settings in which methodological choices have direct consequences for what is estimated, what information can be borrowed, and how uncertainty is reported. In each application, I will examine when distributional change is large enough to alter the appropriate target, weighting scheme, or inferential procedure. My objective is to develop methods whose theoretical guarantees, computational implementation, and empirical targets remain aligned as the underlying data environment changes.

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